

EDUGENIE — SAMPLE PROJECT DOCUMENTATION

Title: EduGenie: Google Gemini Powered Learning Assistant

Objective

Provide an integrated assistant for student questions, explanations, summaries, quizzes and learning recommendations.

Solution

FastAPI web application using Gemini for Q&A, summarization, quizzes and learning recommendations, with local LaMini-Flan-T5-783M for explanation.

Technology

Python, FastAPI, Uvicorn, HTML, CSS, JavaScript, Jinja2, Gemini API, Transformers, PyTorch, pytest, GitHub Actions.

Testing and security

Automated tests plus browser testing; credentials supplied through environment variables and excluded from Git.

Future

Voice, multilingual support, gamification and adaptive learning.